

Neeraj Agrawal

Bangalore, India -560037, +917353202027

neeraj13788@gmail.com

<https://in.linkedin.com/in/neeraj-agrawal>

Objective

Demonstrate ability to inspire and mentor data scientists, as well as drive strategic long-term data science initiatives across the organisation.

Education

- **Indian Institute of Science** Bangalore, India
Ph.D in Machine Learning; **CGPA: 9.5/10** Jan 2022 – Present
- **Indian Institute of Science** Bangalore, India
M.E. in Electrical Engineering; **CGPA: 7.3/8; Gold Medalist** Aug 2011 – July 2013
 - **Thesis:** Control and Start-up Diagnostics of Three Phase Inverter;
- **Harcourt Butler Technological Institute** Kanpur, India
B.Tech in Electrical Engineering; **Percentage: 84; 2nd Rank** Aug 2007 – May 2011

Technical Skills

- **AI/ML & GenAI:** Large Language Models, Agentic Frameworks (Google ADK), Vertex AI, Cloud Speech-to-Text (Chirp), RAG, Multimodal In-Context Learning. Deep learning, NLP
- **Cloud & Architecture:** Google Cloud Platform (Cloud Run, Cloud Firestore, Google Cloud Storage, Secret Manager), Serverless Architecture, Microservices.
- **Languages & Frameworks:** Python, Flask, Streamlit, FastAPI, Docker, Terraform.

Work Experience

- **Solution Architect: Gen AI** Bangalore, India
Google Sep 2024 – Present
 - **Led AI Architecture for a Global Product Launch:** Directed the Generative AI strategy for a major electronics manufacturer's new smart display line. Embedded onsite for a month to develop the real-time transcription PoC, resolving critical API bottlenecks just days before launch to secure top innovation awards at ISE 2025.
 - **Closed Strategic Deal with Tier-1 IT Integrator:** Architected a multi-agent AI framework (using Google ADK) to solve complex legal data extraction and policy comparison challenges. The success of this highly specialized system directly resulted in a signed commercial contract.
 - **Architected Enterprise-Scale AI Agents:** Designed and deployed a conversational voice agent for a multinational conglomerate. Built a robust multi-agent framework utilizing enterprise connectors (ServiceNow, Microsoft Entra, Google Drive) to seamlessly orchestrate complex IT workflows and secure data retrieval.
 - **Optimized Enterprise AI Costs (TCO):** Architected scalable, serverless AI infrastructures across multiple enterprise clients. Implemented smart routing between different GenAI models (e.g., Gemini Flash for volume, Pro for reasoning) to drastically reduce running costs while maintaining premium performance.
- **Staff Data Scientist: Voice** Bangalore, India
Walmart Jul 2020 – Sep 2024
 - Leading initiative to fine tune LLM for summarization task which will help customer care agents to resolve users queries faster
 - Leading different projects to build few shot entity and intent classifiers using LLMs
 - Lead/Built contextual NLU project which predicts intents based on users utterance together with user context
 - Created a deep learning based intent prediction model for customer care chatbot. Deployed the model using java play framework to serve high traffic.
 - Built a hierarchical intent prediction model to predict granular intents which helps in increase in automation rate.
- **Senior Data Scientist: Personalization** Bangalore, India
Walmart Jan 2019 – Jul 2020

- Built a machine learning model to generate item pairs that can be bundled together based on user data.
- Developed a personalised model to increase CTR (Click Through Rate) in transactional email recommendations.
- Developed a personalised recommendation model to recommend better items in promotional emails.

- **Data Scientist: Sentience Data Analytics Team** Bangalore, India
Honeywell Technology Solution Oct 2016 – Jan 2019
 - Developed a deep learning model to identify faults in utility poles using images captured by drones.
 - Developed an analytic model to predict health of AHUs, that helped in preventing unscheduled maintenance and downtime.
 - Developed an analytic model to identify the anomaly in manufacturing process of glass films.
- **Domain Analytic Engineer: Prognostics and Analytics Team** Bangalore, India
GE Global Research Center & GE Transportation Jul 2013– Oct 2016
 - Developed and implemented a machine learning model fused with domain knowledge to predict the locomotive battery failure.
 - Developed an analytical model to predict the failure of Auxiliary and Traction Alternator in locomotive.
 - Analyzed, developed and validated predictive model for Air Compressor Leakage System.
 - Developed a prescriptive model for maintenance of Traction Motor prognostics.
 - Developed the experimental test-bed consisting of three power inverters emulating UPS units connected in the IP ring bus architecture, to test the control algorithm.

Mentoring and Research Experience

- **PhD Scholar in IISc** IISc Bangalore, India
Neeraj Agrawal Jan 2022 – Present
 - Developed few-shot prompting strategies using In-Context Learning for unseen tasks, achieving state-of-the-art results
- **Mentor @Springboard** Springboard India, Bangalore
Neeraj Agrawal Mar 2019 – Jan 2021
 - Mentored more than 10 mentees as part of springboard's data science program and guided them in transitioning their career in data science.
- **Control And Startup Diagnostics of Three Phase Inverter** IISc Bangalore
Neeraj Agrawal, Vinod John Aug 2012 – July 2013
 - Developed and validated on inverter hardware an algorithm that can detect fault in LCL filter, DC bus capacitor and IGBTs before starting of three phase inverter.

Publications and Patents

- **Publications**
 - N. Agrawal, and S. Ganapathy, "Spoken language understanding on unseen tasks with in-context learning." In Interspeech 2025.
 - N. Rajput, H. B. Sandhibigraha, N. Agrawal and V. M. Iyer, "An Empirical Model Informed Neural Network Core Loss Predictor for Soft Magnetic Materials," in IEEE Transactions on Power Electronics, vol. 40, no. 8, pp. 11257-11267, Aug. 2025
 - M. Chen et al., "MagNet Challenge for Data-Driven Power Magnetics Modeling," in IEEE Open Journal of Power Electronics, vol. 6, pp. 883-898, 2025
 - S. Nandi, N. Agrawal, "Improving Few-Shot Cross-Domain Named Entity Recognition by Instruction Tuning a Word-Embedding based Retrieval Augmented Large Language Model" In Proceedings of the Conference on Empirical Methods in Natural Language Processing: Industry Track, Nov 2024
 - S. Nandi, N. Agrawal et al., "Enhancing Customer Service Chatbots with Context-Aware NLU through Selective Attention and Multi-task Learning", In CODS-COMAD, Dec. 2024
 - N. Agrawal, S. Kumar, P. Bhatt, T. Agarwal, "Hierarchical Text Classification Using Contrastive Learning Informed Path Guided Hierarchy," In ECAI 2023, pp. 19-26. IOS Press, 2023
 - S. Kumar, S. Bansal, N. Agrawal, P. Bhatt, "Building a Few-Shot Cross-Domain Multilingual NLU Model for Customer Care," In ECAI 2023, pp. 3212 - 3217. IOS Press, 2023
 - N. Agrawal and V. John, "Enhanced Startup Diagnostics of LCL Filter for an Active Front-End Converter," Journal of Power Electronics, vol. 18, no. 5, pp. 1567-1576, 2018. doi: 10.6113/JPE.2018.18.5.1567.

- N. Agrawal and V. John, "LCL filter startup diagnostics for an Active Front End converter," 2016 IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES), 2016, pp. 1-6, doi: 10.1109/PEDES.2016.7914471.
- N. Agrawal, P. Paritosh, D. Jyoti, "Optimal Health Care Inventory Management Using Analytics", Proceedings of the International Conference on Database, Data Warehouse, Data Mining and Big Data (DDDDBD2015), Jakarta, Indonesia 2015, pp. 37-42, ISBN: 978-1 -941968-20-8 ©2015 SDIWC2

• US Patents

- Nandi, Subhadip; Agrawal, Neeraj; Modak, Sudipta; 2025. System and method for few-shot cross-domain named entity recognition. U.S. Patent application 8789US01/8789US02, filed 2025. Patent pending
- Nandi, Subhadip; Agrawal, Neeraj; 2025. Predicting Intent Using Context Aware Natural Language Understanding Model. U.S. Patent application 7007US01, filed 2025. Patent pending
- Agrawal, Neeraj; Bhatt, Priyanka; Agarwal, Tanishka. 2022. Systems and Methods for Generating a Customized Graphical User Interface. U.S. Patent 12,112,127, filed Jan. 31, 2022. and issue Oct. 8, 2024
- Agrawal, Neeraj; Singh, Anshika; Bhatt, Priyanka. 2022. Automatically Determining User Intent by Sequence Classification Based on Non-Time-Series-Based Machine Learning. U.S. Patent application 17/589552, filed Jan. 31, 2022. Patent pending
- Agrawal, Neeraj; Bhatt, Priyanka. 2021. Systems and methods for utilising feedback data. U.S. Patent 12,229,784, filed Jun. 30, 2021. issue Feb. 18, 2025
- Agrawal, Neeraj; Pati, Dibyajyoti; Paritosh, Prakarsh. 2021. Route examination system. U.S. Patent 10,940,876, filed Jan. 3, 2018 and issued Mar. 9, 2021.
- Smith, Mark; Agrawal, Neeraj. 2020. System and method for locomotive diagnostics, U.S. Patent 10,661,816, filed Sep. 18, 2018 and issued May 26, 2020.
- Tiwari, Arvind; Kolhatkar, Yashomani; Agrawal, Neeraj. 2018. Hybrid ac and dc distribution system and method of use. U.S. Patent 9,859,716, filed May 29, 2015 and issued Jan. 2, 2018.

• Thought Leadership & Blog Publications

- **Google Cloud Blog:** "How to build a real-time voice agent with Gemini and Google ADK" (Aug 2025) – Codified scalable architectural patterns for multi-modal streaming, low-latency bidirectional communication, and real-time enterprise data retrieval.
- **Google Cloud Blog:** "How to build a deep research agent for lead generation using Google's ADK" (Aug 2025) – Architected a hierarchical, parallelized multi-agent system capable of dynamic reasoning and advanced state management to automate complex analytical workflows.
- **Google Cloud Blog:** "A guide to converting ADK agents with MCP to the A2A framework" (Jul 2025) – Authored a step-by-step engineering guide for upgrading standalone ADK agents with Model Context Protocol (MCP) tools into fully interoperable Agent-to-Agent (A2A) ecosystems.
- **Google Cloud Blog:** "How to build a simple multi-agentic system using Google's ADK" (Jul 2025) – Established best practices for scaling enterprise AI by replacing brittle, monolithic LLMs with specialized, high-fidelity multi-agent teams.
- **Medium:** "An AI Travel Agent in Action: A Detailed Look at How Two Agents Plan a Trip" (Jun 2025) – Demonstrated practical, collaborative task orchestration and delegation between distinct AI agents within a unified framework.

Certifications

- **Google Cloud Professional Machine Learning Engineer** (May 2025)
- **Google Cloud CEPF L300 Architect Training** (Nov 2025)

Awards/Honors Achievements

- **Google Cloud Excellence / Spot Award:** Awarded for exceptional technical leadership in architecting a multi-agent AI framework that directly secured a major commercial contract with a Tier-1 Global IT Integrator.
- **Winner Walmart Global Techathon (2019, 2021,2021, 2022, 2023):** Won multiple global hackathons in Walmart
- **Above and Beyond Achievement(2014):** Award for successful execution of project on UPS power sharing in GRC.
- **N R Khambhati Memorial Gold Medal(2013):** By the council of Indian Institute of Science (IISc) as Best ME Student in Electrical Engineering program.